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Blood Cancer Identification using Hybrid Ensemble Deep Learning Technique

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***Abstract*—Blood cancer-related illnesses may be challenging to examine and diagnose, both of which can take a significant period of time. During the course of the preceding decade, a variety of methods for detecting, analysing, and categorizing blood cancer in people were established. These methods include: However, as of right now, there is neither a method nor a model available that can automate the process of inspecting human blood cells to determine whether or not cancer is present. A model with these characteristics has the potential to improve disease detection and prevention, which would lead to a prompter medical diagnosis. This study presents the research progress made in the area of Deep Learning Model to identify the abnormalities in blood cells. In order to detect blood cancer, this study has used Hybrid Ensemble Deep Learning method. This method results with an accuracy of more than 95%.**

Keywords—Stress Detection, Electrocardiogram (ECG), Brain-Computer Interface (BCI), Electroencephalogram (EEG), Phonocardiogram (PCG)

I. INTRODUCTION

Disease progression can be tracked by monitoring the blood changes. Leukemia may be lethal if it is not treated [1]. According to statistics, leukaemia is the sixth highest cause of death for both women and men. When leukaemia initially manifests in the body, it does so in the bone marrow [2].

There is a tiny tissue inside of every bone that is also called bone marrow (see Figure 1.1). Only by studying the white blood cells may leukaemia be discovered [3]. Here have thus limited Here investigation to the white blood cells (WBCs).

Bone marrow cells begin to transform into leukaemia or infected cells after contracting an infection. There seems to be something unusual about these leukaemia cells compared to healthy ones [4]. Compared to typical cells, they develop abnormally and live for longer than usual. In order to do perform this task, they disrupt healthy cells. While healthy cells perish after a specific period of time, leukaemia cells persist [5]. Leukaemia is characterised by the abnormal survival and proliferation of both old and new cells. Different types of leukaemia have varying rates of cell progression [6].

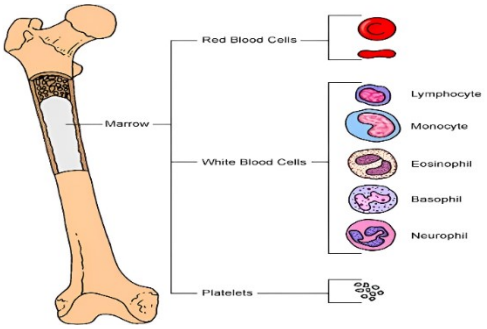

Figure 1. What's in your bones

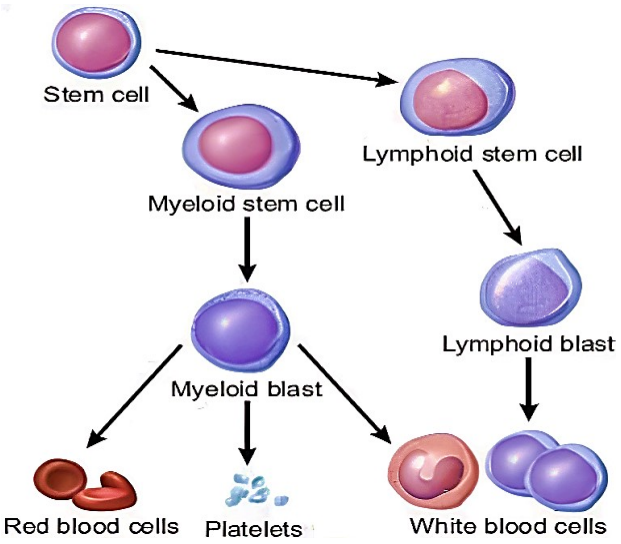

Figure 2. Progress in stem cell research

Figure 2 shows that the stem cells may split off into either myeloid or lymphoid stem cells [11]. In order to examine microscopic pictures of human blood cells for signs of leukaemia, the Google Net deep transfer architecture, which is often used in the learning process, is applied here [12]. The approach that has been presented is able to differentiate between normal and the four different kinds of leukemia—acute, lymphocytic, myeloid, and chronic—using microscopic pictures of human blood samples as the source material [13].

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There are five types of white blood cells in the body, and they are called lymphocytes, myelocytes, neutrophils, basophils, and eosinophils. When the bone marrow produces abnormally large numbers of white blood cells, it is called leukaemia [14].

There should be a natural decline in the number of these aberrant white blood cells over time, but they don't, leading to an inflated total. There are so many aberrant white blood cells that they prevent the healthy cells from executing their job [15].

II. Literature Survey

Demir et al [1], Leukocytes are cells generated in the bone marrow that account for around 1% of total blood cells. Blood cancer develops when these white blood cells proliferate uncontrollably. The SN-AM dataset is used in the proposed research to categorise Acute Lymphoblastic Leukemia (ALL) and Multiple Myeloma (MM), two of the three cancer types. If your bone marrow produces an abnormally high number of lymphocytes, you may develop acute lymphoblastic leukaemia (ALL), a kind of cancer. Multiple myeloma (MM), on the other hand, causes malignant cells to cluster in the bone marrow rather than spreading throughout the body. They also impede the body from generating new, healthy blood cells because they take up so much room. It used to take a long time and required the expertise of a professional. Using deep learning methodologies, the suggested strategy decreases the chance of human mistakes.

Ranjitha et al [2], Leukemia is a kind of blood cancer that is one of the major causes of mortality in the United States. In the case of white blood cells, which are both visible and in their early phases, microscopic imaging enables a range of cancer diagnostic procedures. The capacity to promptly and efficiently detect leukaemia is crucial in assisting doctors in giving patients with the necessary therapy. The segmentation step starts with the separation of white blood cells from other blood components such as erythrocytes and platelets using statistical factors such as mean and standard deviation. The area and perimeter of white blood cell nuclei were examined for their potential value in leukaemia diagnosis. K-means is used in this suggested technique for identifying cancer at an early stage.

Reza et al [3], In this paper, the authors recommend using MNSs early on to identify anomalies. Consider the detection of cancer cells in a specific portion of a blood artery to set the scene. Biomarkers are chemicals produced and released by cancer cells that serve as diagnostic indicators in the circulatory system. Because cancer biomarkers are often present in extremely low concentrations in early-stage cancer samples, finding them with normal blood testing may be challenging. Cancer biomarkers, on the other hand, are highly concentrated in close proximity to malignant cells. When placed in close proximity to cancer cells, a sensor particularly developed to detect these markers may aid in the diagnosis of cancer.

Siraj et al [4], TML and DL techniques for MIA are thoroughly discussed, with an emphasis on leukocyte categorization in blood smear pictures and other medical imaging domains such as MRI, CT images, X-rays, and ultrasounds. The primary contribution of the proposed review is to identify the best TML and DL algorithms in MIA, particularly for leukocyte categorization in blood smear pictures. This study offers a comprehensive look at existing DL techniques, with a focus on developing convolutional neural network-based models in the MIA domain. According to studies in this field, typical TML processes are often employed to analyse microscopic blood smear images for white blood cells (WBC). They help in the detection of haematological disorders like AIDS and blood cancer, and the information they give is priceless to medical specialists. AIDS and blood cancer (Leukemia) are two instances, and the measures are detailed here.

Tusneem et al [5], The recommended strategy is divided into two stages: Following the extraction of a ROI using the CMYK-moment localization approach, deep learning-based features are constructed utilising a CNN-based feature fusion process. Finally, the ROI's correctness is assessed. Many different categorization techniques are utilised when judging the value of the information collected. The twelve distinct feature extraction strategies were evaluated on an external dataset and compared to other methods in a benchmarking exercise.

Dharani et al [6], The abnormal rise in the number of white blood cells in the bone marrow causes leukaemia, often known as blood cancer. The predominant kind of white blood cells may be used to define disease progression. Acute and chronic leukaemia are the two main types of this disease. Leukaemia is classified into subtypes based on the cell lineages that are affected. Lymphocytic/lymphoid leukaemia is caused by the malignant transformation of lymphocytes, while myeloid/myelocytic leukaemia is differentiated by the production of myelocytes, granulocytes, and other myeloid cells as well as white blood cells. As a consequence, acute and chronic leukaemia are classified as lymphoid (ALL), myeloid (AML), lymphocytic (CLL), and myeloid (CML) (CML). The SVM classifier is a kind of machine learning classifier that may be used to categorise data. This article examines blood smear photographs from healthy and leukemic individuals, using image processing techniques to distinguish between different forms of blood cancer.

Omer et al [7], The Google Net deep transfer architecture, which is used in learning, is used to search for signs of leukaemia in microscopic pictures of human blood cells. From microscopic pictures of human blood samples, the suggested approach can identify between normal and the four kinds of leukemia—acute, lymphocytic, myeloid, and chronic.

III. Methodology

The proposed method suggests an automated approach for diagnosing blood cancer by using two Mixed Neural Networks (MNNs) and transfer learning to identify key features in each image. The two MNN models used in the

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method are Multi-Layer Perceptron (MLP) and Convolutional Neural Network (CNN).

MNNs are regularized versions of multilayer perceptron, where each neuron in one layer is connected to all neurons in the next layer. the fully connected networks makes them prone to overfitting. Typical ways of regularization include adding some form of magnitude measurement of weights to the loss function. Therefore, on the scale of accuracy and complexity, MNNs are on lower extreme. There are four layers which involved in MNN which are Convolutional layer to convolve the input and pass the result to next layer, Pooling layer to progressively reduce the spatial size of the representation to reduce the amount of parameters and computation in network, Flattening layer for converting the data into all dimensional array for putting it to the next layer and then Fully connected layer which connect every neuron in one layer to every neuron in another layer.

By incorporating data from several abstraction levels that may be considered supplementary characteristics, here technique is able to improve classification accuracy beyond what was previously feasible. Integrating feature maps at higher dimensions might potentially boost the discriminative strength of intermediate features and sidestep the problem of network gradient.

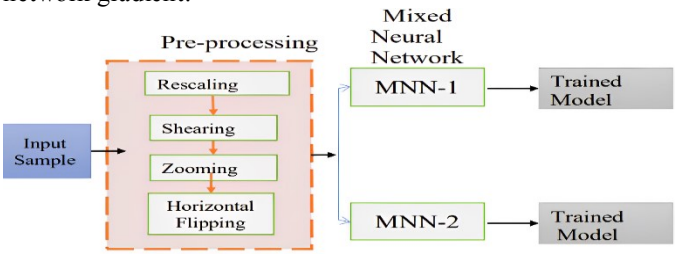

Figure 4. Approaches to Training

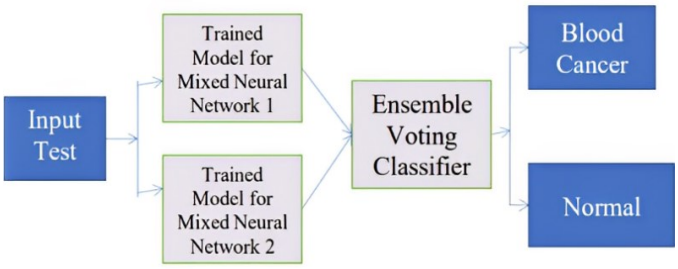

Figure 3. Approaches to Testing

A. Dataset collection:

The dataset on blood cancer includes both healthy people and those who have been diagnosed with blood cancer.

The information included in this assemblage was obtained from the Kaggle website. It is an online resource that is devoted to data science, and it gives us access to medical imaging data.

B. Preprocessing:

The raw input training data is subjected to a variety of data pre-processing procedures, including zooming, shearing, flipping, and rescaling, amongst others.

Using the shear tool, one side of an image, layer, selection, or path can be moved in a specific direction while simultaneously flipping the other side. This tool can also be used to create curved edges. When zooming in an image, more pixels are added so that the overall size of the image may be increased.

The most difficult part of this process is interpolating the new pixels using the pixels that were there before in the surrounding region.

Flipping a picture either horizontally or vertically may result in a mirror effect being formed.

For example, after applying a horizontal flip, the pixel that was formerly positioned at (x, y) will now be found at (width - x -1), y).

C. Training Process:

The data is rescaled in every dimension of the spectrum using a technique called rescaling. After that, the Mixed neural network model is fed the information (MNN). Each layer of the MNN—convolutional, polling, flattening, and fully connected—contributes to the model's accuracy.

D. Testing:

After applying the two training models to the test data, the final prediction was generated using an Ensemble voting Classifier. This came after the application of the training models. The conclusion reaches a conclusion on the presence or absence of blood cancer in the case.

IV. Ensemble Voting Classifier

Combining machine learning classifiers that are either conceptually similar to one another or wholly dissimilar to one another is what the Ensemble Vote Classifier does in order to accomplish classification by majority or plurality vote.

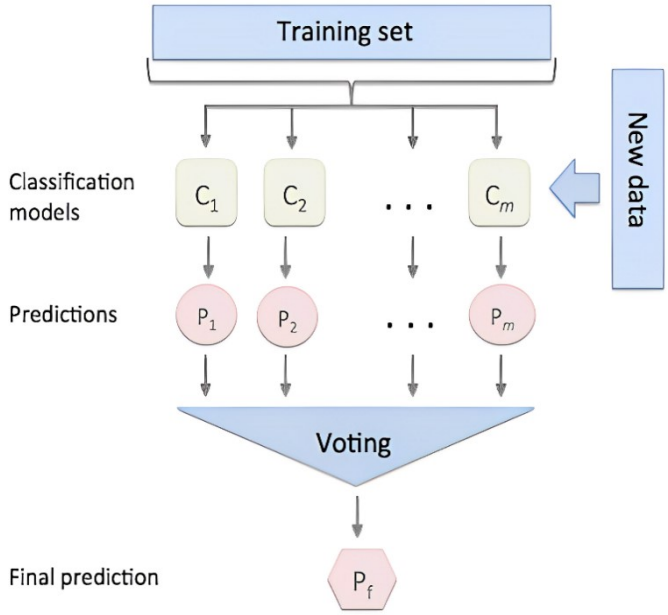

Figure 4. Ensemble Vote Classifier:
A majority voting classifier

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A technique to data classification known as the Ensemble Vote Classifier allows for both hard and soft voting to be conducted. If there is a close vote, the final class label is often the class label that has been predicted the majority of the time by the classification models.

This is because the classification models have more data to work with. Predictions of class labels are arrived at by the method of soft voting, which involves assessing the probability of each class (only recommended if the classifiers are well calibrated).

It would be quite good to have some sense of the degree of precision with which expect being supplied with the outcome when it is eventually delivered.

A. Hard voting:

A majority vote might be as easy as a hard vote. Here a majority (plurality) vote from all classifiers CjCj to determine the predicted class label i^:

$$\hat{i}=\text{mode}\{H1(j),\ H2(j)...,\ Hm(j)\}$$

If Herewere to combine three different classification systems, then one training sample would be categorised as follows:

- classifier no. 1 ->area 0
- classifier no. 2 ->area 0
- classifier no. 3 ->area 1

$$\hat{i}=\text{mode}\ \{0,0,1\}\ =0$$

The sample has been designated as "area 0" according to the votes of the majority.

B. Proportional Representation Vote:

By assigning a weight wj to the corresponding classifier Cj, Heremay calculate a weighted majority vote in addition to the simple majority vote (hard voting) mentioned in the preceding section.

$$\hat{y}=\arg\max\sum_{j=1}^{m} w_j \chi_A(C_j(x)=i)$$

where χA is the characteristic function $[C_j(x)=i\in A]$, and AA is theset of unique class labels.

Continuing with the example from the previous section

- classifier 1 ->area 0
- classifier 2 ->area 0
- classifier 3 ->area 1

assigning the weights {0.2, 0.2, 0.6} would yield a prediction y^=1:

$$\arg\max[0.2\times i0+0.2\times i0+0.6\times i1]=\ 1$$
$$1\arg\max[0.2\times i0+0.2\times i0+0.6\times i1]=1$$

C. Soft Vote:

Predicted probabilities pp for classifier are used in soft voting to make predictions about the class labels.

$$\hat{y}=\arg\max\sum_{j=1}^{m} w_j p_{ij},$$

$$\hat{y}=\arg\max\sum_{j=1}^{m} w_j p_{ij},$$

Consider the preceding section to have been a class labelled binary classification problem. h∈ {0,1

- H1(x)→[0.3,0.2]
- H2(x)→ [0.5,0.4]
- H3(x)→[0.7,0.6]

The average probability is calculated using uniform weights:

$$\hat{y}=\arg\max[p(i0|x),\ \ p(i1|x)]\ =0$$
$$\hat{y}=\arg\max[p(i0|x),p(i1|x)]=0$$
$$\hat{y}=\arg\max[p(i0|x),p(i1|x)]\ =1$$

IV. Experiments and Results

The results of the constructed model are discussed and compared in a number of experimental settings and presented in this part. As a cherry on top, the proposed model is implemented in Python language using the Keras module for deep learning framework, and all recreations are run on Google CoLab with a 12GB needed in NVIDIA and also Tesla K80 GPU. Two datasets are used in the experiment, and their performance is verified in both isolation and in combination.

This statistic reflects the proportion of all classes that the trained model correctly predicted, such as Acute Lymphoblastic Leukaemia (ALL) vs all other possible diseases. This metric shows the ratio of those diagnosed with leukaemia to those who do not have it. As accuracy increases, the model's precision improves [68-71].

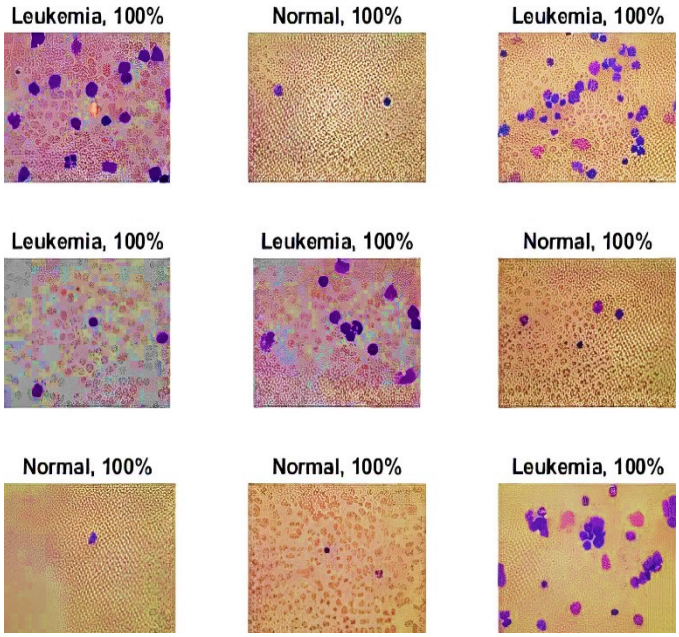

Figure 5. Ensemble Vote Classifier: majority votingclassifier

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V. RESULT

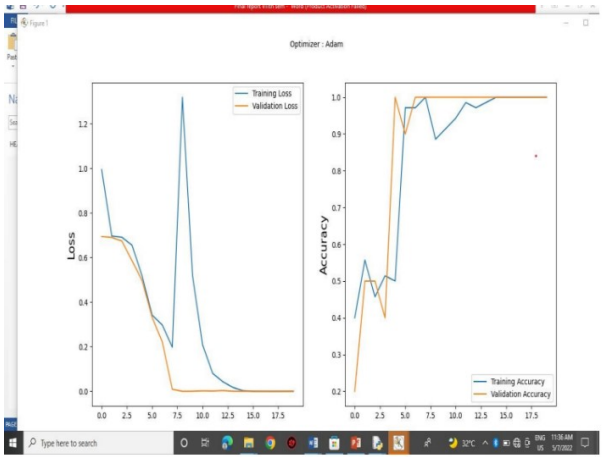

Fig 6. Training Set 1

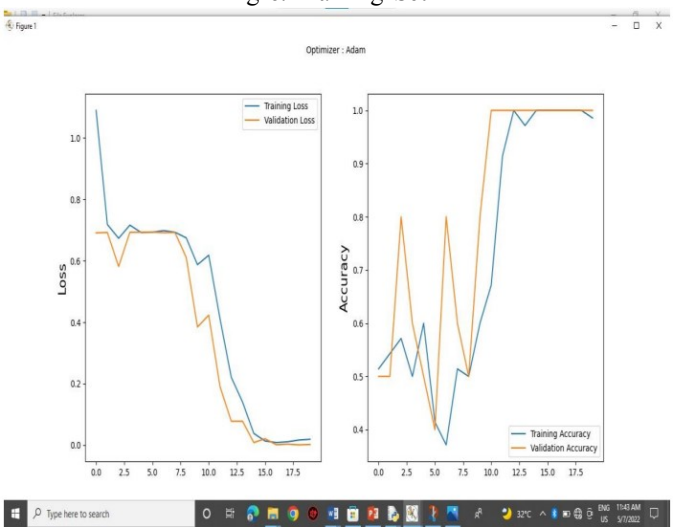

Fig 7. Training Set 2

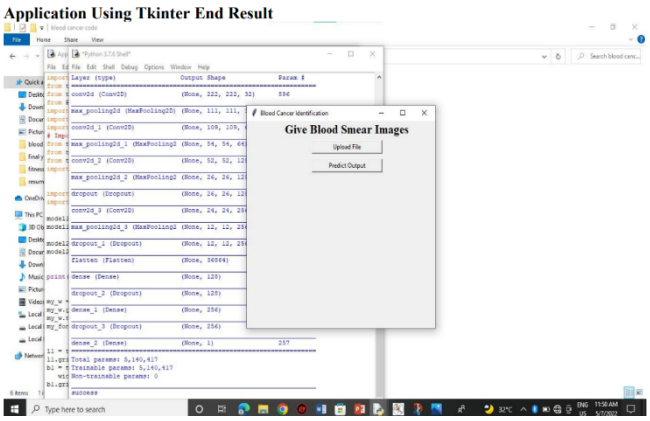

Fig 8. Results of a Tkinter-Based Application

V. CONCLUSION AND FUTURE WORK

The number of individuals whose cancer has spread to their blood contributes significantly to the overall cancer death rate, making it one of the most important factors. In recent years, a number of studies have been conducted to provide deep learning-based approaches for detecting blood cancer. One of these methods is transfer learning strategies. The findings of these studies show an extraordinary level of precision. Despite this, a sizable portion of the academic community continues to invest heavily in the research and development of deep learning algorithms. As a result, deep learning model based neural network detects blood cancer from a very small sample of a patient's blood. Testing on publicly available datasets revealed that the proposed model is more accurate than the existing methods.

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